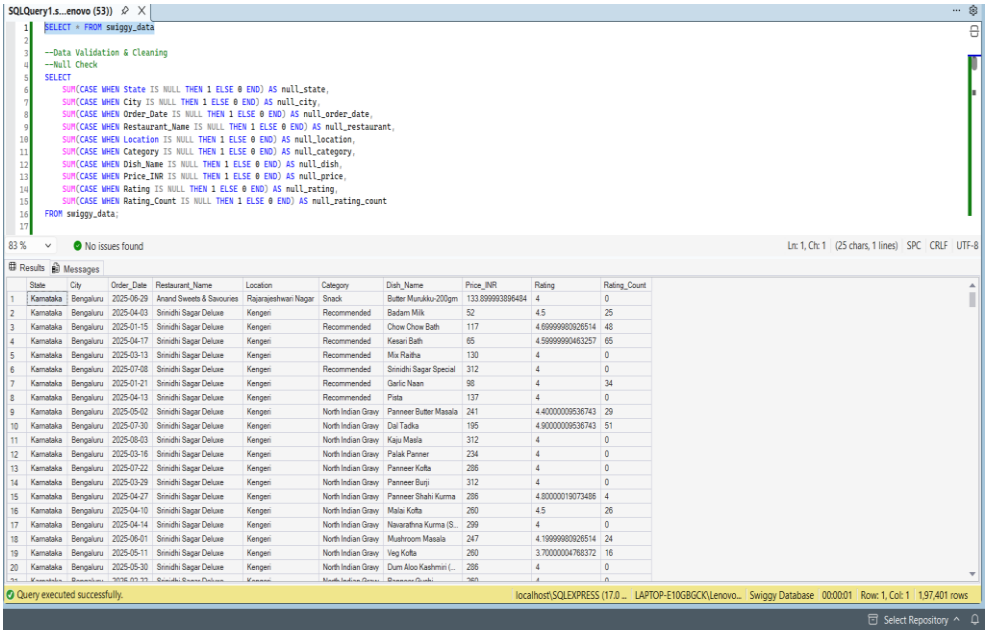
SWIGGY SQL SALES ANALYSIS PROJECT

SELECT * FROM swiggy_data



	State	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count
1	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.899993896484	4	0
2	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52	4.5	25
3	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117	4.69999980926514	48
4	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65	4.59999990463257	65
5	Karnataka	Bengaluru	2025-03-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mix Raitha	130	4	0
6	Karnataka	Bengaluru	2025-07-08	Srinidhi Sagar Deluxe	Kengeri	Recommended	Srinidhi Sagar Special	312	4	0
7	Karnataka	Bengaluru	2025-01-21	Srinidhi Sagar Deluxe	Kengeri	Recommended	Garlic Naan	98	4	34
8	Karnataka	Bengaluru	2025-04-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Pista	137	4	0
9	Karnataka	Bengaluru	2025-05-02	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Butter Masala	241	4.40000009536743	29
10	Karnataka	Bengaluru	2025-07-30	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Dal Tadka	195	4.90000009536743	51
11	Karnataka	Bengaluru	2025-08-03	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Kaju Masla	312	4	0
12	Karnataka	Bengaluru	2025-03-16	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Palak Panner	234	4	0
13	Karnataka	Bengaluru	2025-07-22	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Kofta	286	4	0
14	Karnataka	Bengaluru	2025-03-29	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Burji	312	4	0
15	Karnataka	Bengaluru	2025-04-27	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Shahi Kurma	286	4.80000019073486	4
16	Karnataka	Bengaluru	2025-04-10	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Malai Kofta	260	4.5	26
17	Karnataka	Bengaluru	2025-04-14	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Navarathna Kurma (S...	299	4	0
18	Karnataka	Bengaluru	2025-06-01	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Mushroom Masala	247	4.19999980926514	24
19	Karnataka	Bengaluru	2025-05-11	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Veg Kofta	260	3.70000004768372	16
20	Karnataka	Bengaluru	2025-05-30	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Dum Aloo Kashmiri (...	286	4	0

1.) DATA CLEANING

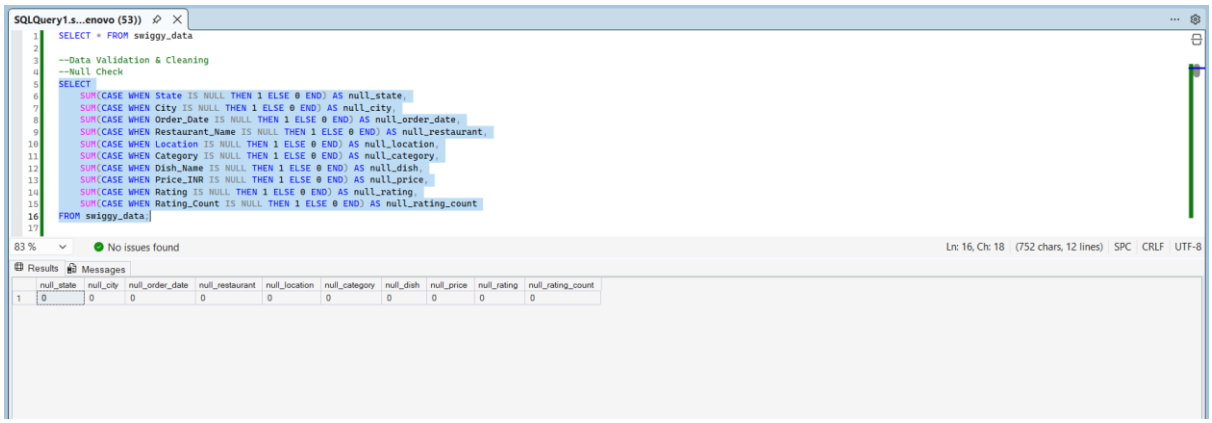
CHECK FOR NULL VALUES IN EACH COLUMN

--Data Validation & Cleaning

--Null Check

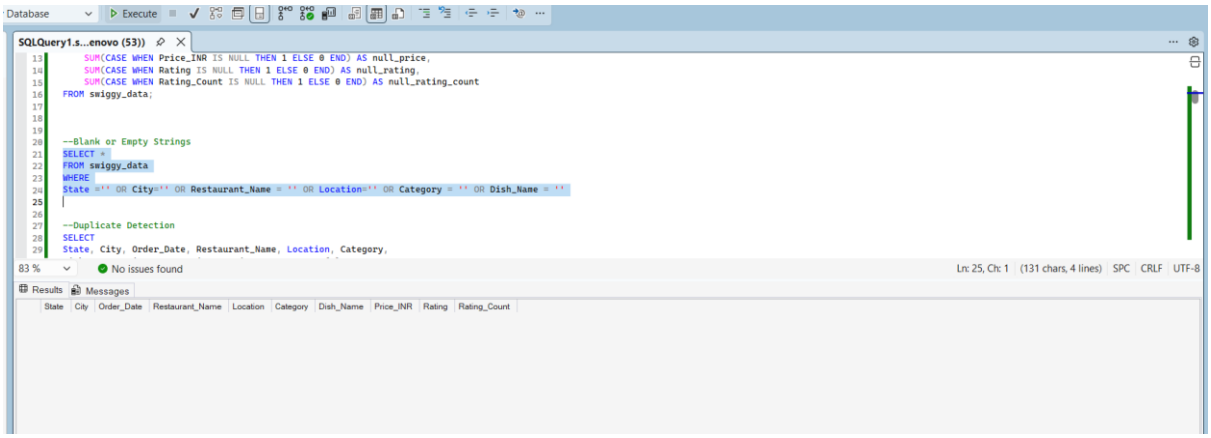
```
SELECT
   SUM(CASE WHEN State IS NULL THEN 1 ELSE 0 END) AS null_state,
   SUM(CASE WHEN City IS NULL THEN 1 ELSE 0 END) AS null_city,
   SUM(CASE WHEN Order_Date IS NULL THEN 1 ELSE 0 END) AS null_order_date,
   SUM(CASE WHEN Restaurant_Name IS NULL THEN 1 ELSE 0 END) AS null_restaurant,
   SUM(CASE WHEN Location IS NULL THEN 1 ELSE 0 END) AS null_location,
   SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS null_category,
   SUM(CASE WHEN Dish_Name IS NULL THEN 1 ELSE 0 END) AS null_dish,
   SUM(CASE WHEN Price_INR IS NULL THEN 1 ELSE 0 END) AS null_price,
   SUM(CASE WHEN Rating IS NULL THEN 1 ELSE 0 END) AS null_rating,
   SUM(CASE WHEN Rating_Count IS NULL THEN 1 ELSE 0 END) AS null_rating_count
FROM swiggy_data;
```

SWIGGY SQL SALES ANALYSIS PROJECT



--BLANK OR EMPTY STRINGS

SELECT *
FROM swiggy_data
WHERE
State =' ' OR City' ' OR Restaurant_Name = ' ' OR Location=' ' OR Category = ' ' OR Dish_Name = ' '



--DUPLICATE DETECTION

SELECT
State, City, Order_Date, Restaurant_Name, Location, Category,
Dish_Name, Price_INR, Rating, Rating_Count, count(*) as CNT
FROM swiggy_data
GROUP BY
State, City, Order_Date, Restaurant_Name, Location, Category,
Dish_Name, Price_INR, Rating, Rating_Count
HAVING count(*)>1

SWIGGY SQL SALES ANALYSIS PROJECT

```
--Delete Duplication
WITH CTE AS (
SELECT *, ROW_NUMBER() Over(
   PARTITION BY State, City, Order_Date, Restaurant_Name, Location, Category,
   Dish_Name, Price_INR, Rating, Rating_Count
ORDER BY (SELECT NULL)
)AS rn
FROM swiggy_data
)
DELETE FROM CTE WHERE rn>1
```

2.) DATA MODELING

```
--CREATING SCHEMA
--DIMENSION TABLES
--DATE TABLE
CREATE TABLE dim_date (
   date_id INT IDENTITY(1,1) PRIMARY KEY,
   Full_Date DATE,
   Year INT,
   Month INT,
   Month_Name varchar(20),
   Quarter INT,
   Day INT,
   Week INT
   )

SELECT * FROM dim_date

--Dim_location
CREATE TABLE dim_location (
   location_id INT IDENTITY(1,1) PRIMARY KEY,
   State VARCHAR(100),
   City VARCHAR(100),
   Location VARCHAR(200)
);

SELECT * FROM dim_location

--Dim_restaurant
CREATE TABLE dim_restaurant (
   restaurant_id INT IDENTITY(1,1) PRIMARY KEY,
   Restaurant_Name VARCHAR(200)
);
```

SWIGGY SQL SALES ANALYSIS PROJECT

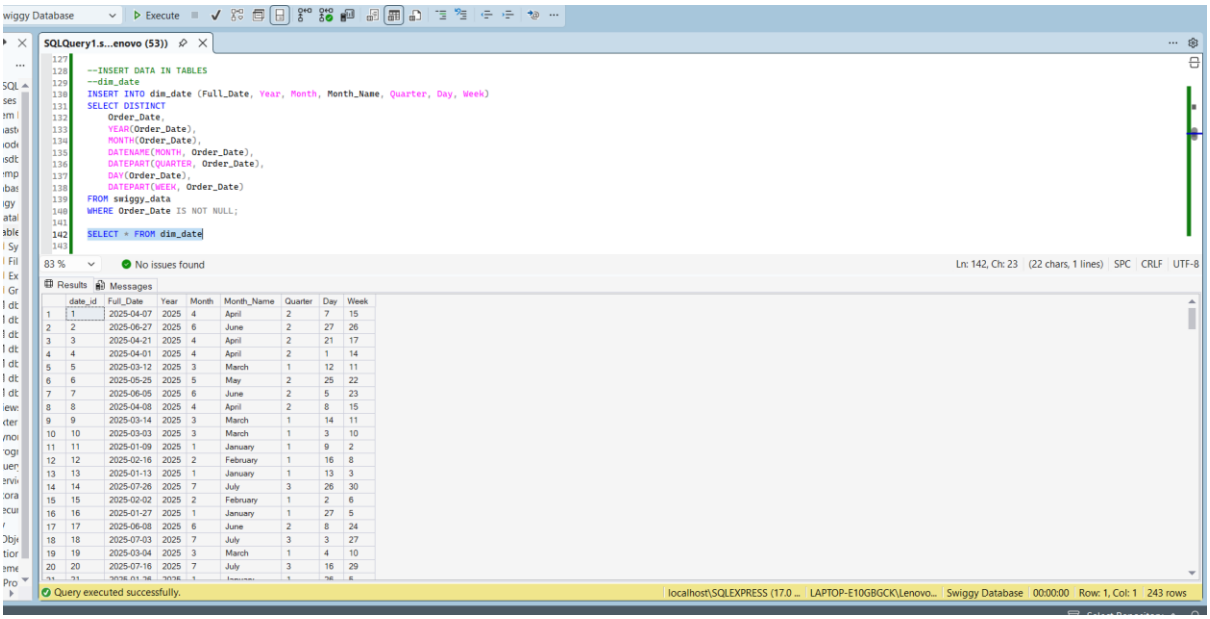
```
--Dim_category
CREATE TABLE dim_category (
   category_id INT IDENTITY(1,1) PRIMARY KEY,
   Category VARCHAR(200)
);


--Dim_dish
CREATE TABLE dim_dish (
   dish_id INT IDENTITY(1,1) PRIMARY KEY,
   Dish_Name VARCHAR(200)
);

SELECT * FROM swiggy_data


--FACT TABLE
CREATE TABLE fact_swiggy_orders (
   oredr_id INT IDENTITY(1,1) PRIMARY KEY,

   date_id INT,
   Price_INR DECIMAL(10,2),
   Rating DECIMAL(4,2),
   Rating_Count INT,

   location_id INT,
   restaurant_id INT,
   category_id INT,
   dish_id INT,

   FOREIGN KEY (date_id) REFERENCES dim_date(date_id),
   FOREIGN KEY (location_id) REFERENCES dim_location(location_id),
   FOREIGN KEY (restaurant_id) REFERENCES dim_restaurant(restaurant_id),
   FOREIGN KEY (category_id) REFERENCES dim_category(category_id),
   FOREIGN KEY (dish_id) REFERENCES dim_dish(dish_id)
);
```

SWIGGY SQL SALES ANALYSIS PROJECT

3.) DATA INSERTION

--**INSERT DATA IN TABLES**
--**dim_date**

```
INSERT INTO dim_date (Full_Date, Year, Month, Month_Name, Quarter, Day, Week)
SELECT DISTINCT
   Order_Date,
   YEAR(Order_Date),
   MONTH(Order_Date),
   DATENAME(MONTH, Order_Date),
   DATEPART(QUARTER, Order_Date),
   DAY(Order_Date),
   DATEPART(WEEK, Order_Date)
FROM swiggy_data
WHERE Order_Date IS NOT NULL;

SELECT * FROM dim_date
```

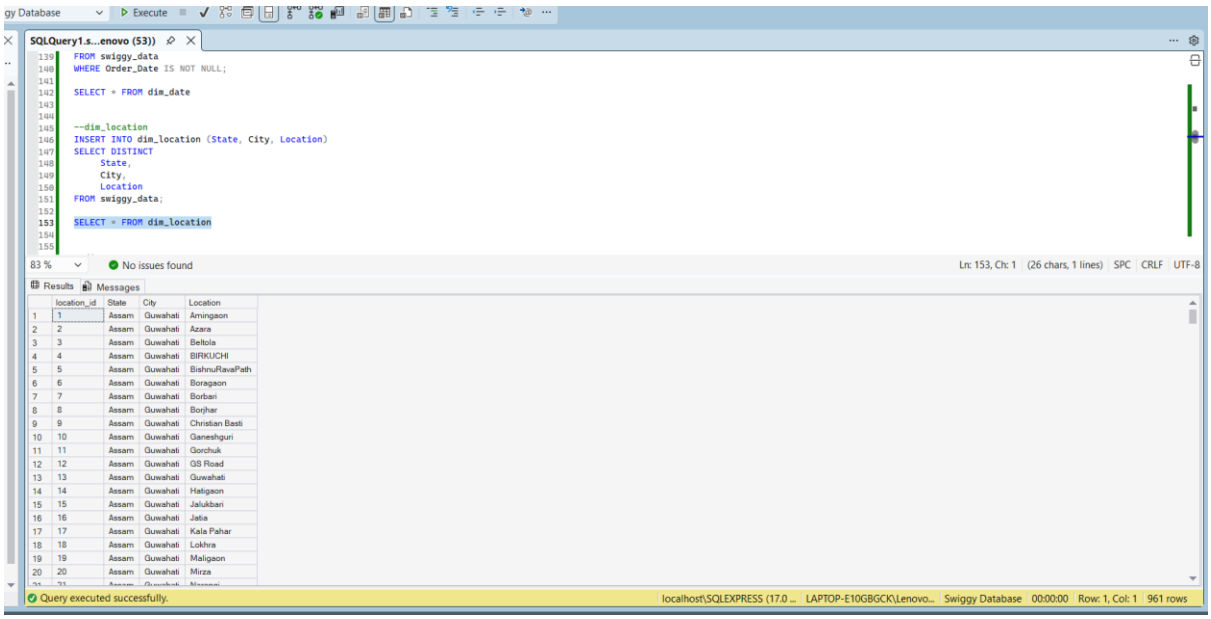


	date_id	Full_Date	Year	Month	Month_Name	Quarter	Day	Week
1	1	2025-04-07	2025	4	April	2	7	15
2	2	2025-06-27	2025	6	June	2	27	26
3	3	2025-04-21	2025	4	April	2	21	17
4	4	2025-04-01	2025	4	April	2	1	14
5	5	2025-03-12	2025	3	March	1	12	11
6	6	2025-05-25	2025	5	May	2	25	22
7	7	2025-06-05	2025	6	June	2	5	23
8	8	2025-04-08	2025	4	April	2	8	15
9	9	2025-03-14	2025	3	March	1	14	11
10	10	2025-03-03	2025	3	March	1	3	10
11	11	2025-01-09	2025	1	January	1	9	2
12	12	2025-02-16	2025	2	February	1	16	8
13	13	2025-01-13	2025	1	January	1	13	3
14	14	2025-07-26	2025	7	July	3	26	30
15	15	2025-02-02	2025	2	February	1	2	6
16	16	2025-01-27	2025	1	January	1	27	5
17	17	2025-06-08	2025	6	June	2	8	24
18	18	2025-07-03	2025	7	July	3	3	27
19	19	2025-03-04	2025	3	March	1	4	10
20	20	2025-07-16	2025	7	July	3	16	29

--**dim_location**

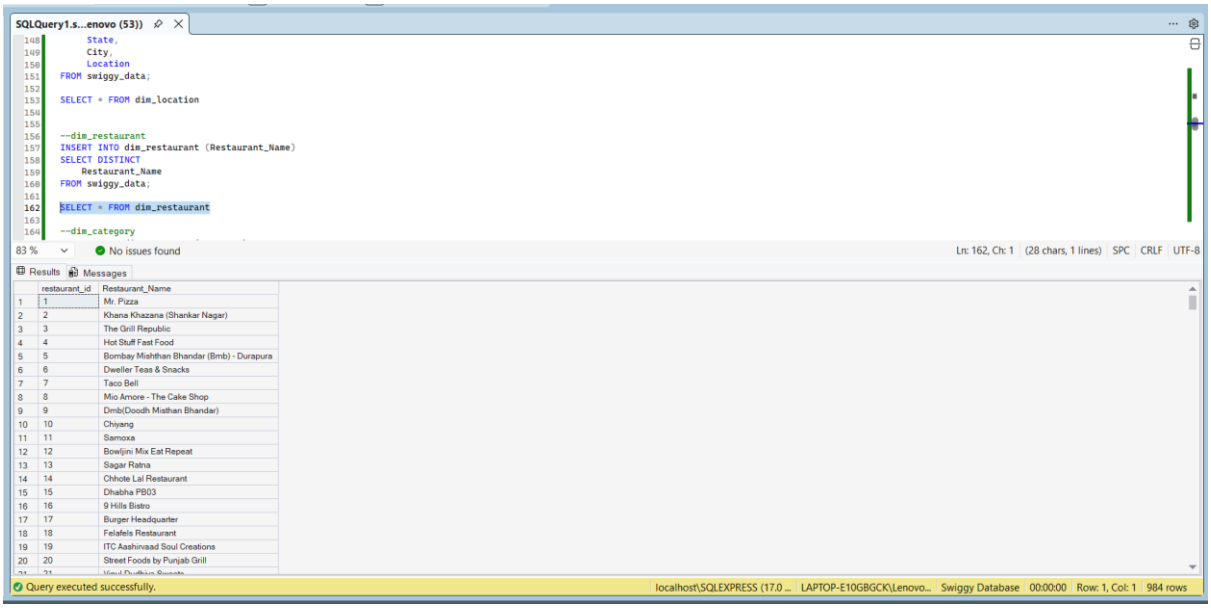
```
INSERT INTO dim_location (State, City, Location)
SELECT DISTINCT
    State,
    City,
    Location
FROM swiggy_data

SELECT * FROM dim_location
```

SWIGGY SQL SALES ANALYSIS PROJECT



--dim_restaurant

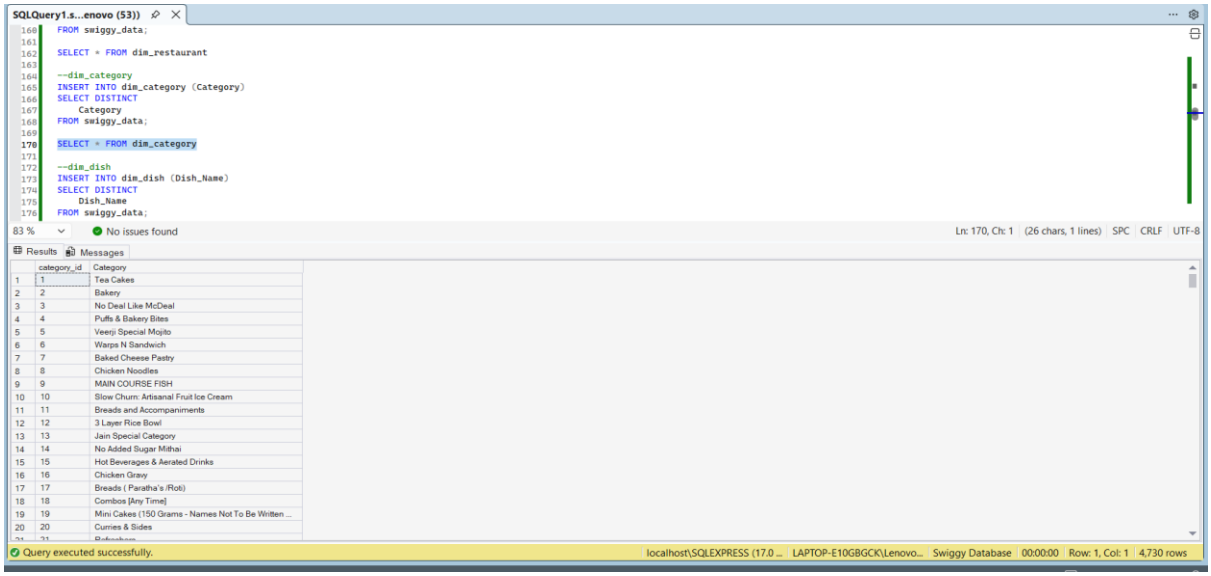
```
INSERT INTO dim_restaurant (Restaurant_Name)
SELECT DISTINCT
    Restaurant_Name
FROM swiggy_data;

SELECT * FROM dim_restaurant
```



SWIGGY SQL SALES ANALYSIS PROJECT

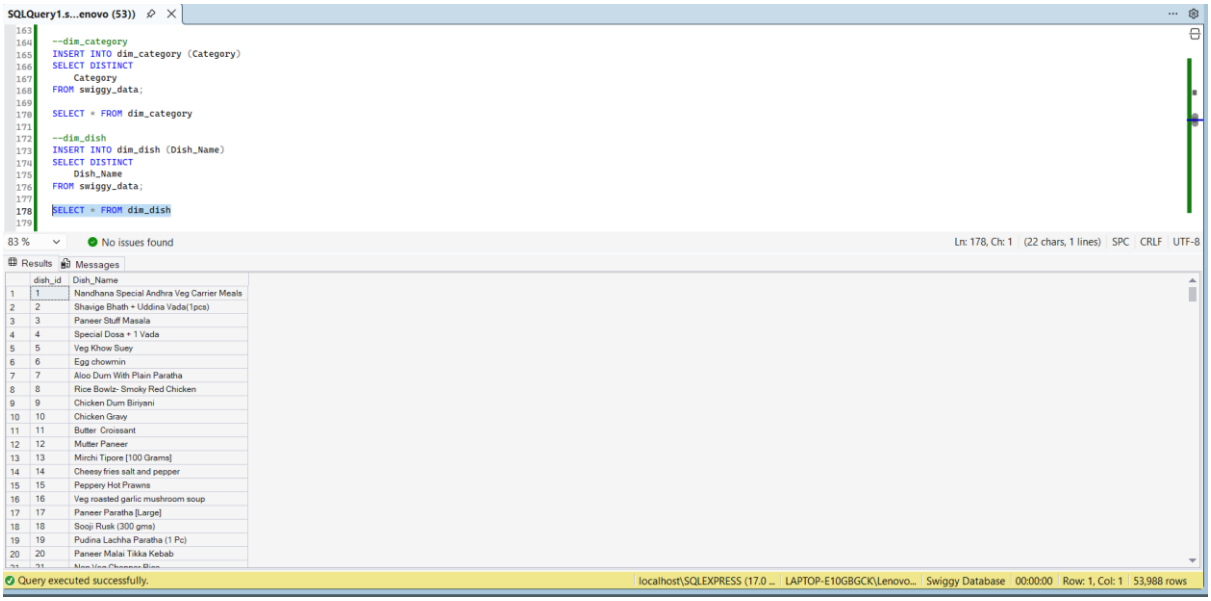
--dim_category

```
INSERT INTO dim_category (Category)
SELECT DISTINCT
   Category
FROM swiggy_data;

SELECT * FROM dim_category
```



--dim_dish

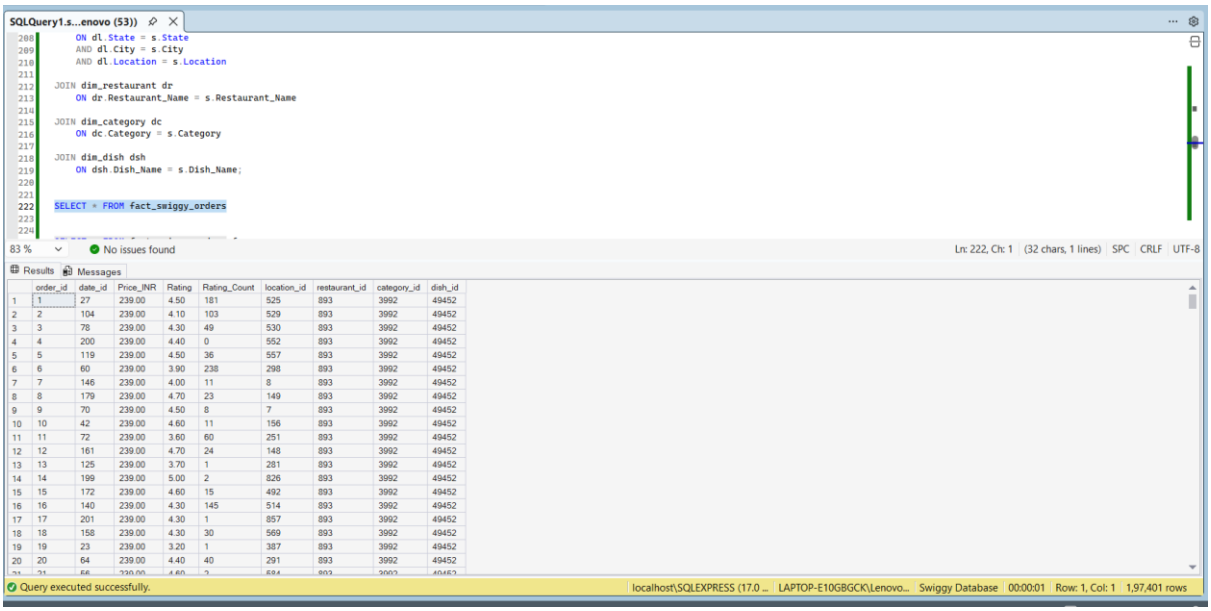
```
INSERT INTO dim_dish (Dish_Name)
SELECT DISTINCT
   Dish_Name
FROM swiggy_data;

SELECT * FROM dim_dish
```



SWIGGY SQL SALES ANALYSIS PROJECT

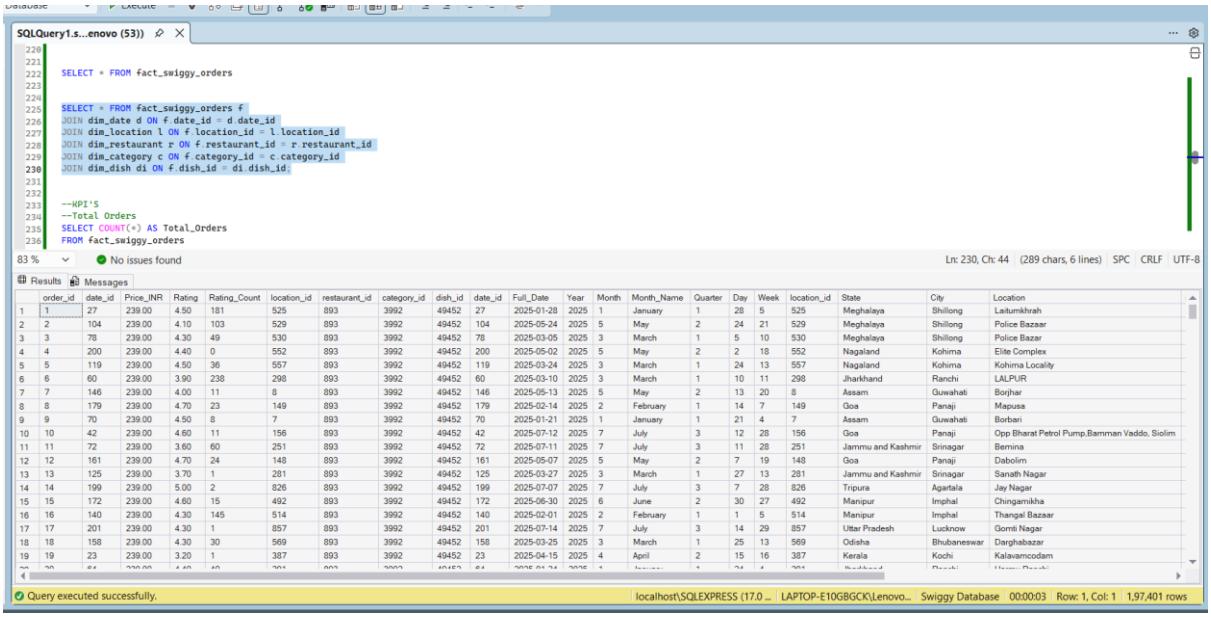
```
--fact_table
INSERT INTO fact_swiggy_orders
(
   date_id,
   Price_INR,
   Rating,
   Rating_Count,
   location_id,
   restaurant_id,
   category_id,
   dish_id
)
SELECT
   dd.date_id,
   s.Price_INR,
   s.Rating,
   s.Rating_Count,

   dl.location_id,
   dr.restaurant_id,
   dc.category_id,
   dsh.dish_id
FROM swiggy_data s

JOIN dim_date dd
   ON dd.Full_Date = s.Order_Date

JOIN dim_location dl
   ON dl.State = s.State
   AND dl.City = s.City
   AND dl.Location = s.Location

JOIN dim_restaurant dr
   ON dr.Restaurant_Name = s.Restaurant_Name

JOIN dim_category dc
   ON dc.Category = s.Category

JOIN dim_dish dsh
   ON dsh.Dish_Name = s.Dish_Name;


SELECT * FROM fact_swiggy_orders
```

SWIGGY SQL SALES ANALYSIS PROJECT



--TOTAL DATA

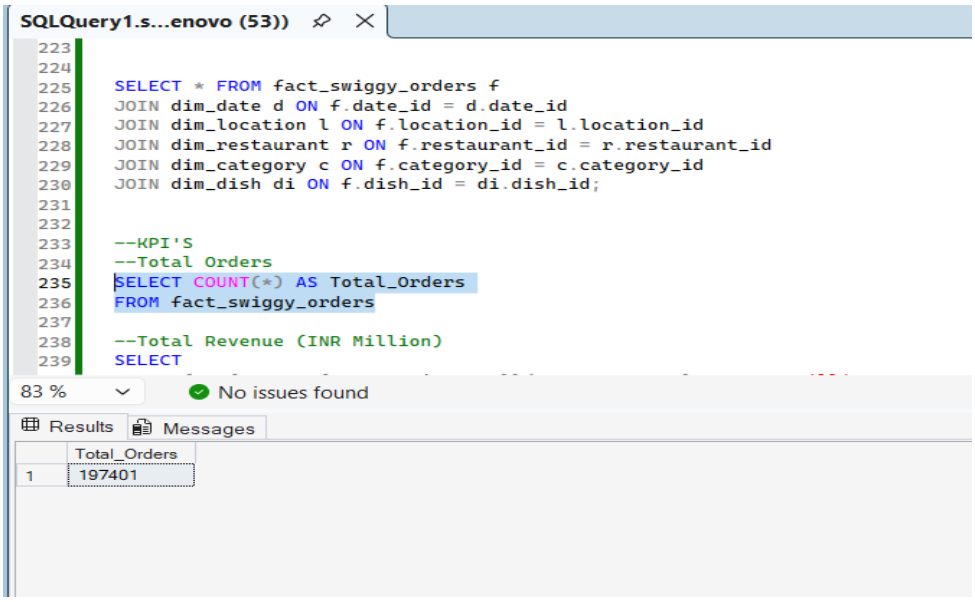
SELECT * FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
JOIN dim_location l ON f.location_id = l.location_id
JOIN dim_restaurant r ON f.restaurant_id = r.restaurant_id
JOIN dim_category c ON f.category_id = c.category_id
JOIN dim_dish di ON f.dish_id = di.dish_id;



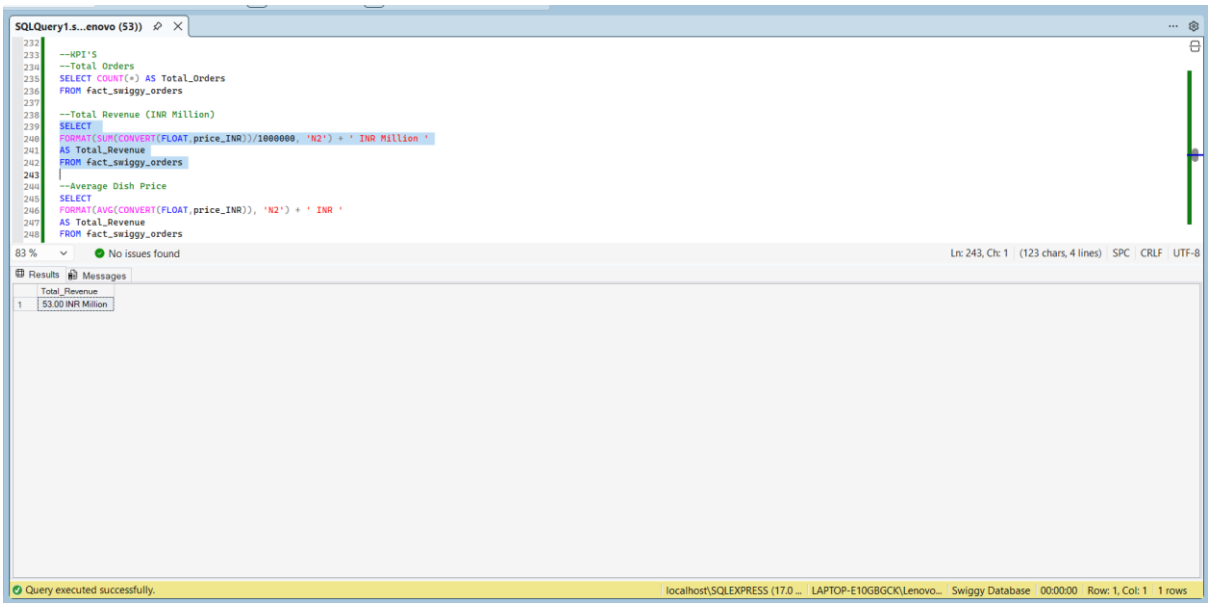
SWIGGY SQL SALES ANALYSIS PROJECT

4.) KPI ANALYSIS

--KPI'S
--Total Orders
SELECT COUNT(*) AS Total_Orders
FROM fact_swiggy_orders



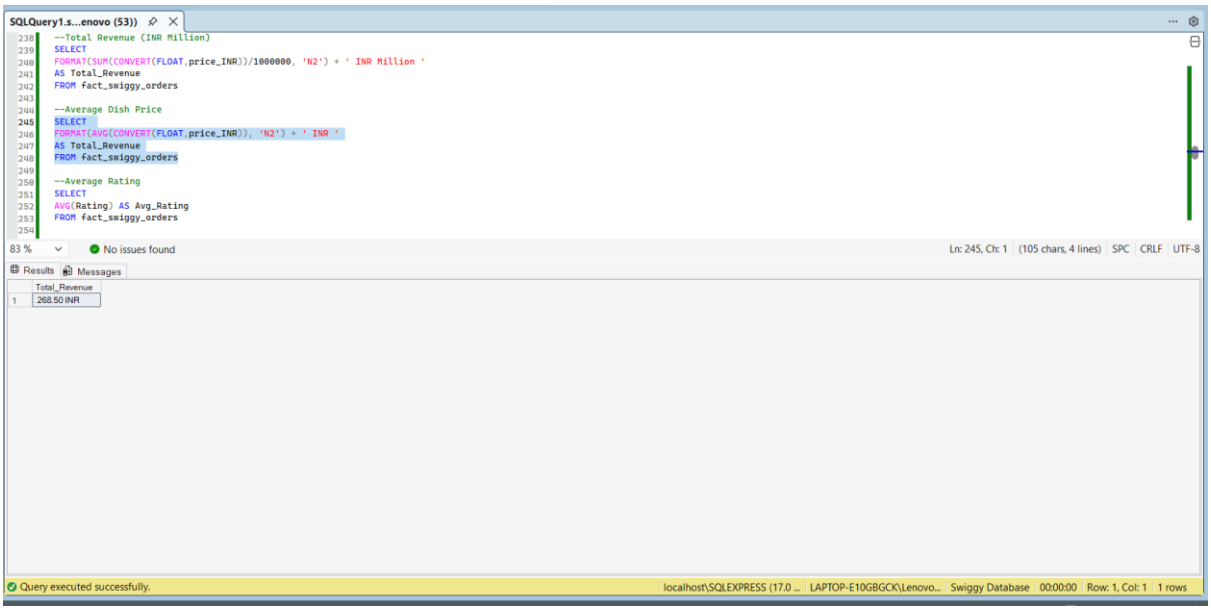
--Total Revenue (INR Million)
SELECT
FORMAT(SUM(CONVERT(FLOAT,price_INR))/1000000, 'N2') + ' INR Million '
AS Total_Revenue
FROM fact_swiggy_orders



SWIGGY SQL SALES ANALYSIS PROJECT

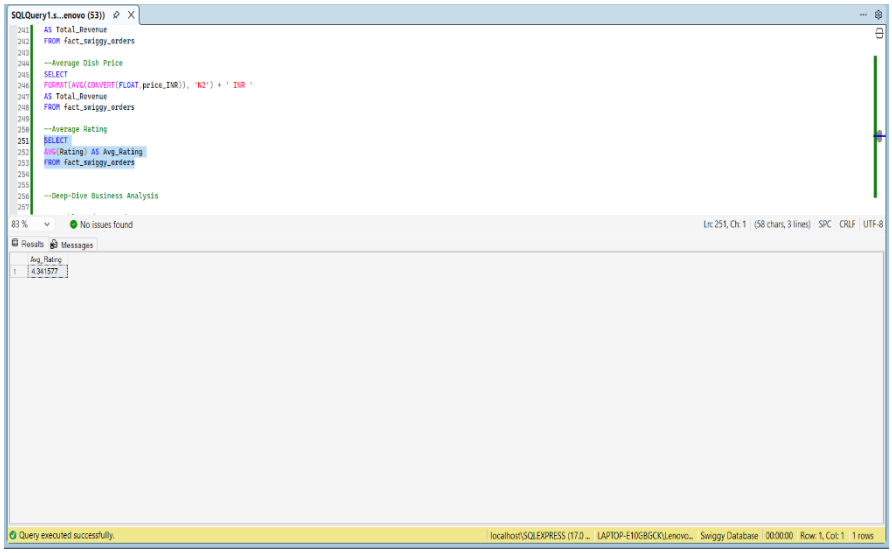
--Average Dish Price

```
SELECT
FORMAT(AVG(CONVERT(FLOAT,price_INR)), 'N2') + ' INR '
AS Total_Revenue
FROM fact_swiggy_orders
```



--Average Rating

```
SELECT
AVG(Rating) AS Avg_Rating
FROM fact_swiggy_orders
```

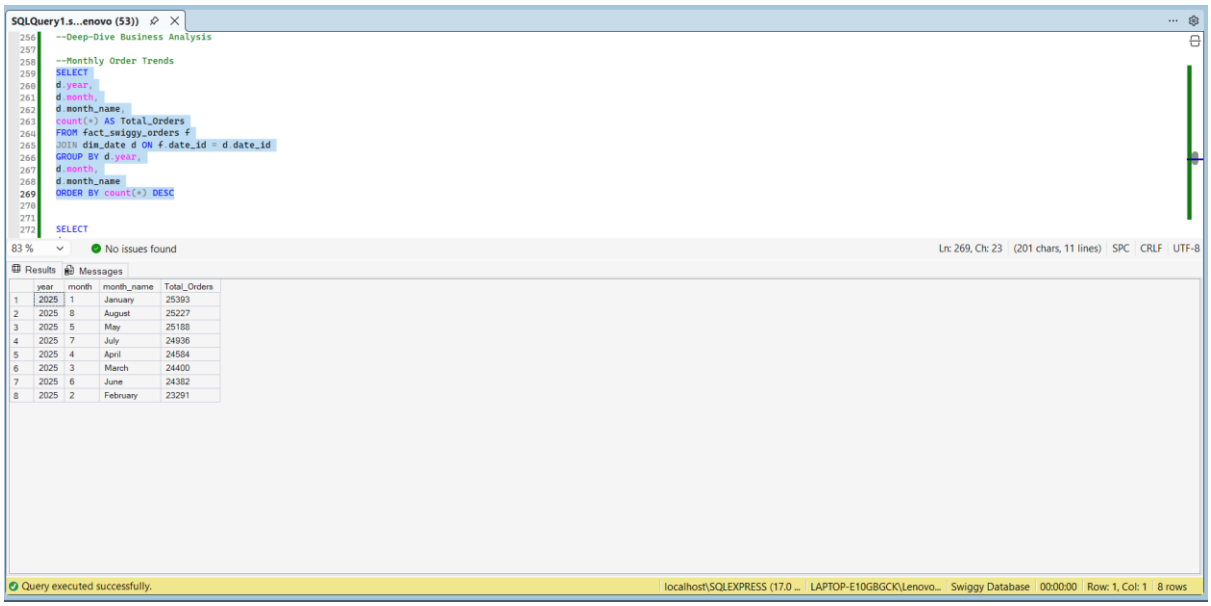


SWIGGY SQL SALES ANALYSIS PROJECT

5.) BUSINESS INSIGHTS

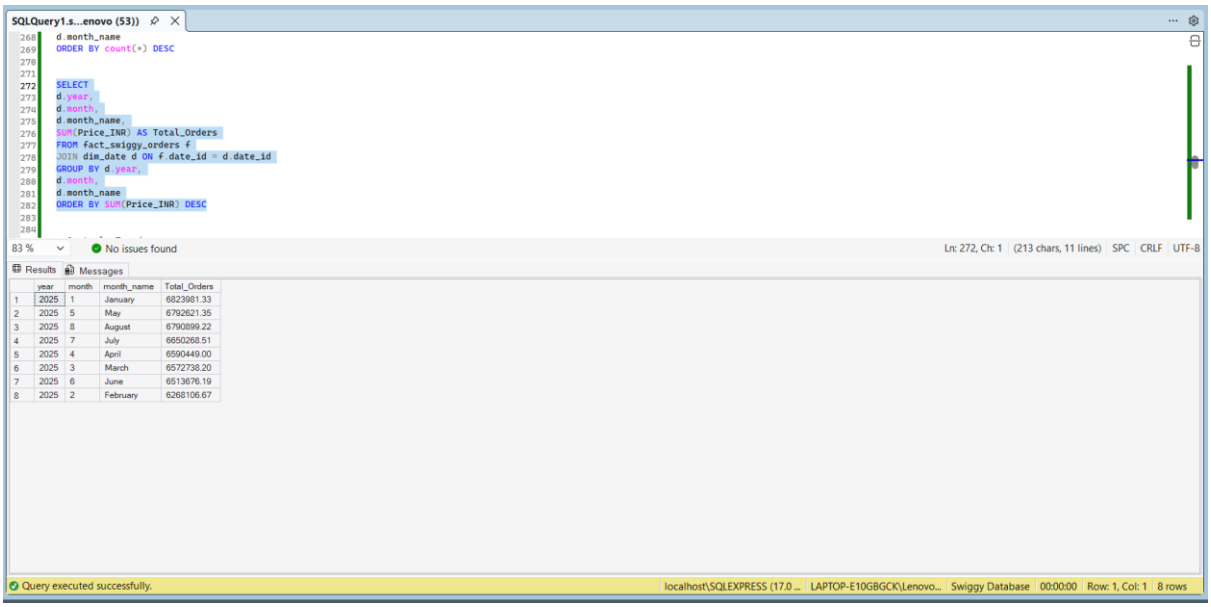
--Deep-Dive Business Analysis
--Monthly Order Trends

```
SELECT
d.year,
d.month,
d.month_name,
count(*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY d.year,
d.month,
d.month_name
ORDER BY count(*) DESC
```



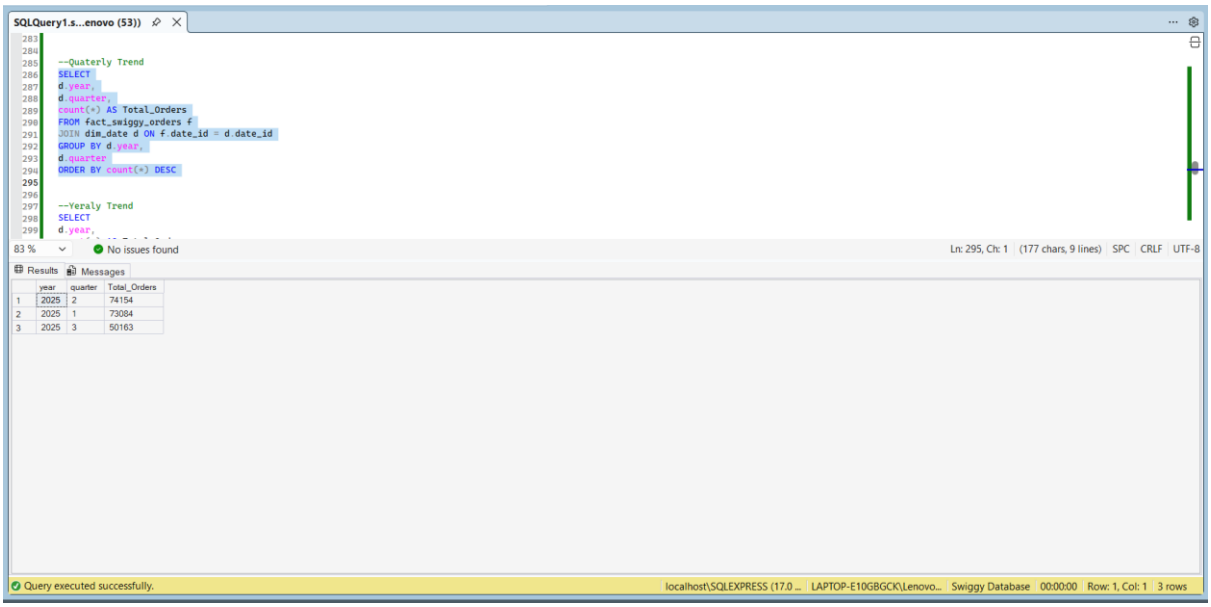

```
SELECT
d.year,
d.month,
d.month_name,
SUM(Price_INR) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY d.year,
d.month,
d.month_name
ORDER BY SUM(Price_INR) DESC
```

SWIGGY SQL SALES ANALYSIS PROJECT



--Quaterly Trend

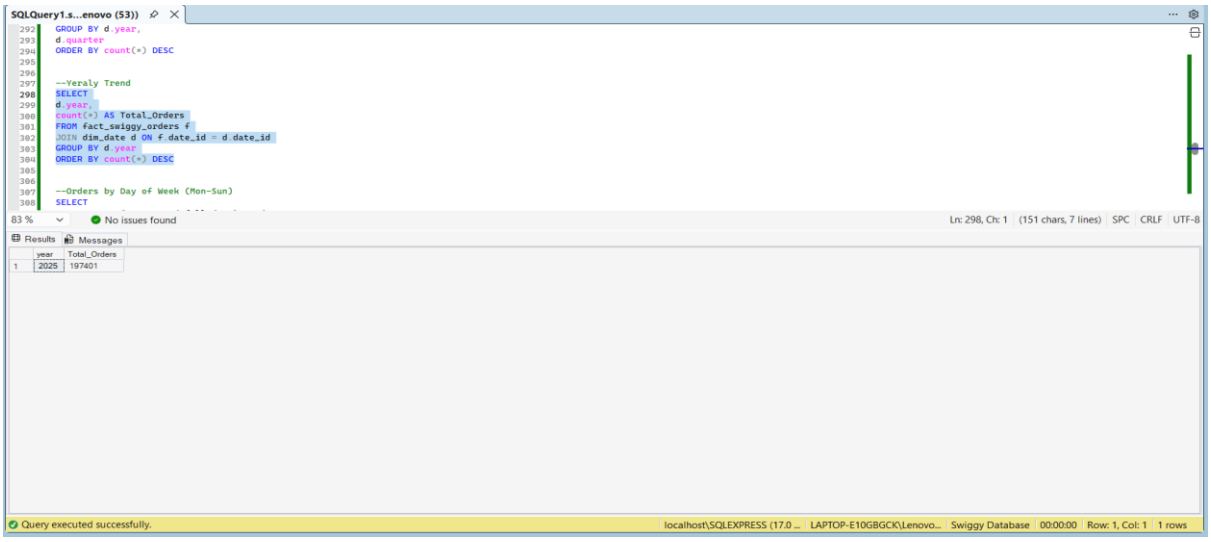
```
SELECT
d.year,
d.quarter,
count(*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY d.year,
d.quarter
ORDER BY count(*) DESC
```



SWIGGY SQL SALES ANALYSIS PROJECT

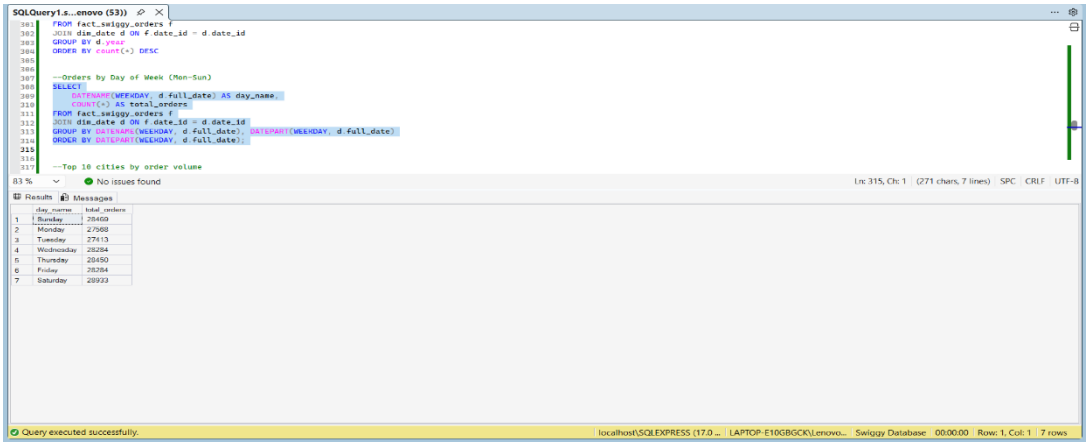
--Yearly Trend

```
SELECT
d.year,
count(*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY d.year
ORDER BY count(*) DESC
```



--Orders by Day of Week (Mon-Sun)

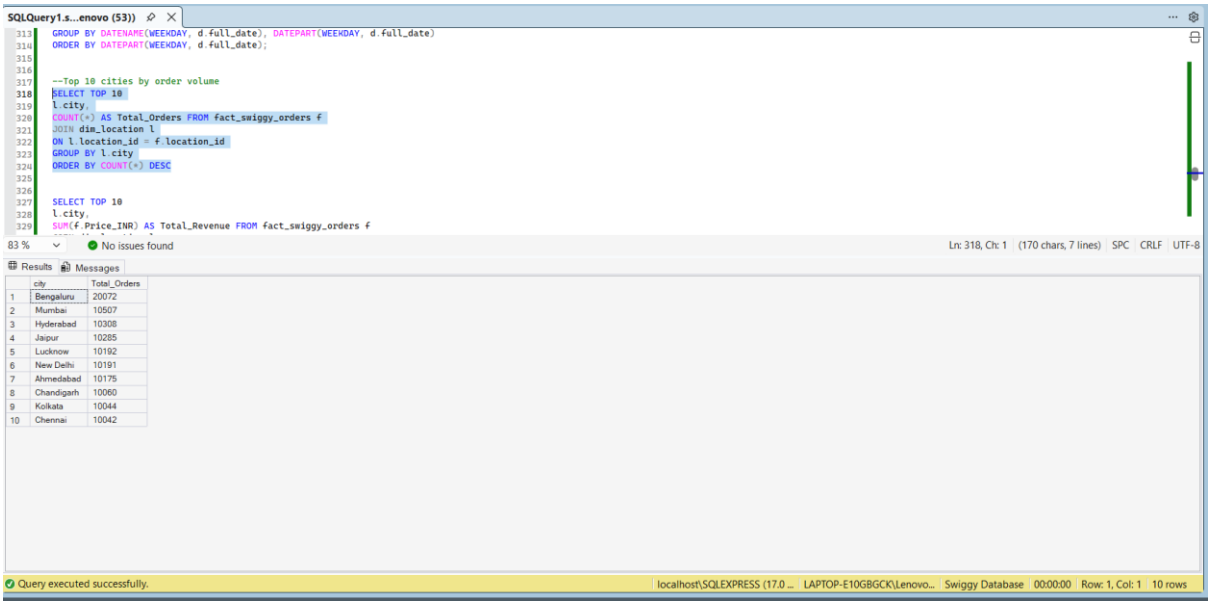
```
SELECT
   DATENAME(WEEKDAY, d.full_date) AS day_name,
   COUNT(*) AS total_orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY DATENAME(WEEKDAY, d.full_date), DATEPART(WEEKDAY, d.full_date)
ORDER BY DATEPART(WEEKDAY, d.full_date);
```



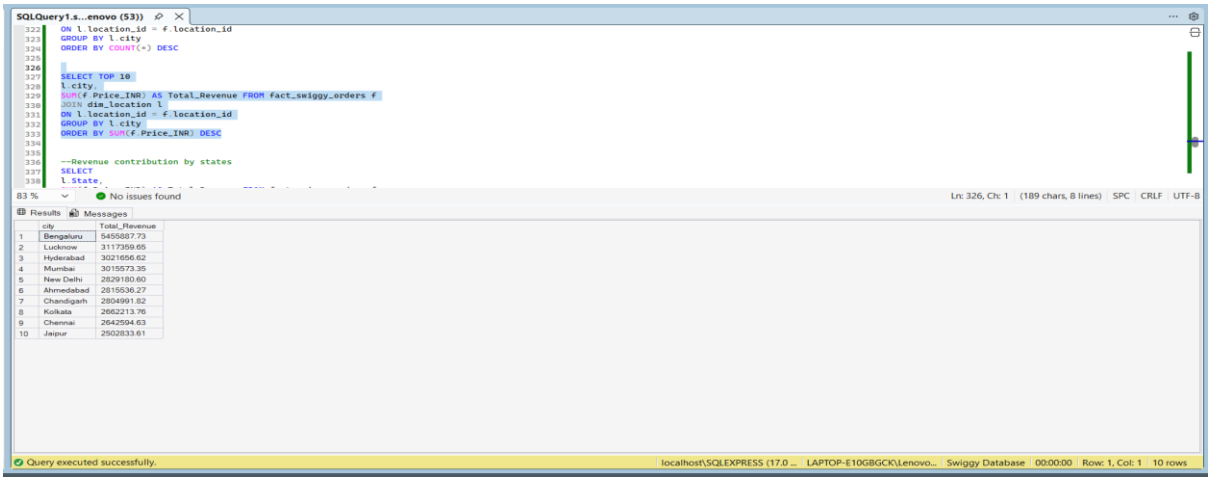
SWIGGY SQL SALES ANALYSIS PROJECT

--Top 10 cities by order volume

```
SELECT TOP 10
l.city,
COUNT(*) AS Total_Orders FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_id
GROUP BY l.city
ORDER BY COUNT(*) DESC
```



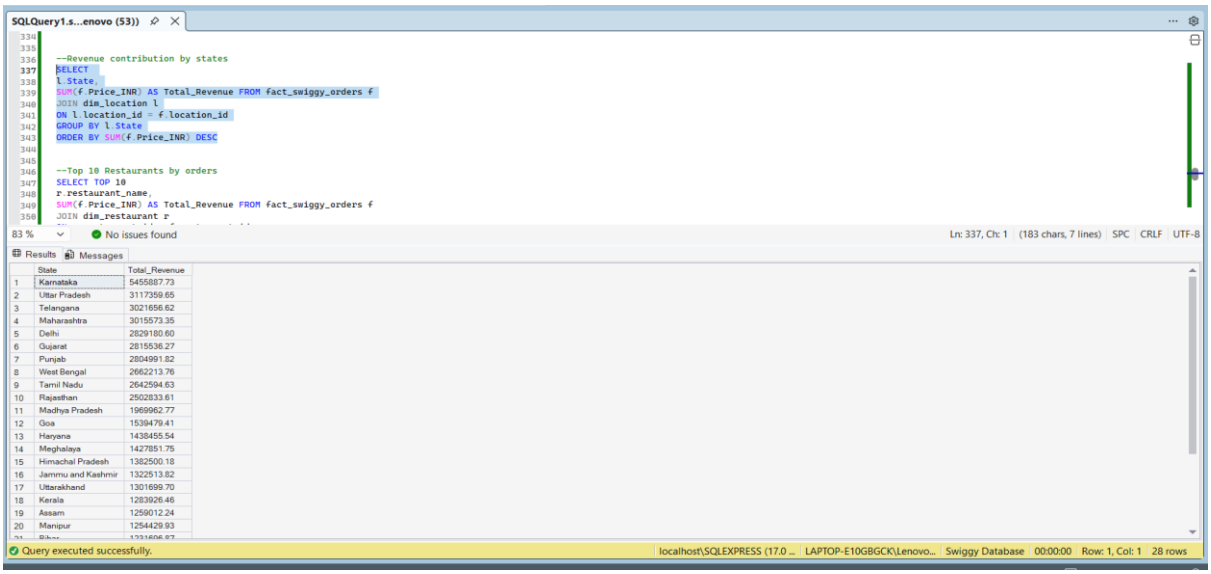

```
SELECT TOP 10
l.city,
SUM(f.Price_INR) AS Total_Revenue FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_id
GROUP BY l.city
ORDER BY SUM(f.Price_INR) DESC
```



SWIGGY SQL SALES ANALYSIS PROJECT

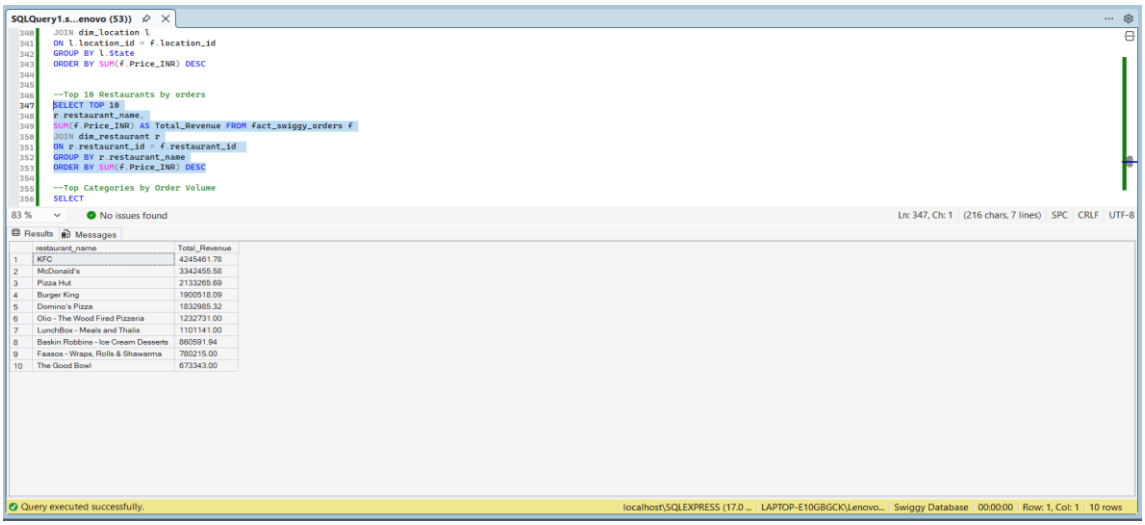
--Revenue contribution by states

```
SELECT
l.State,
SUM(f.Price_INR) AS Total_Revenue FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_id
GROUP BY l.State
ORDER BY SUM(f.Price_INR) DESC
```



--Top 10 Restaurants by orders

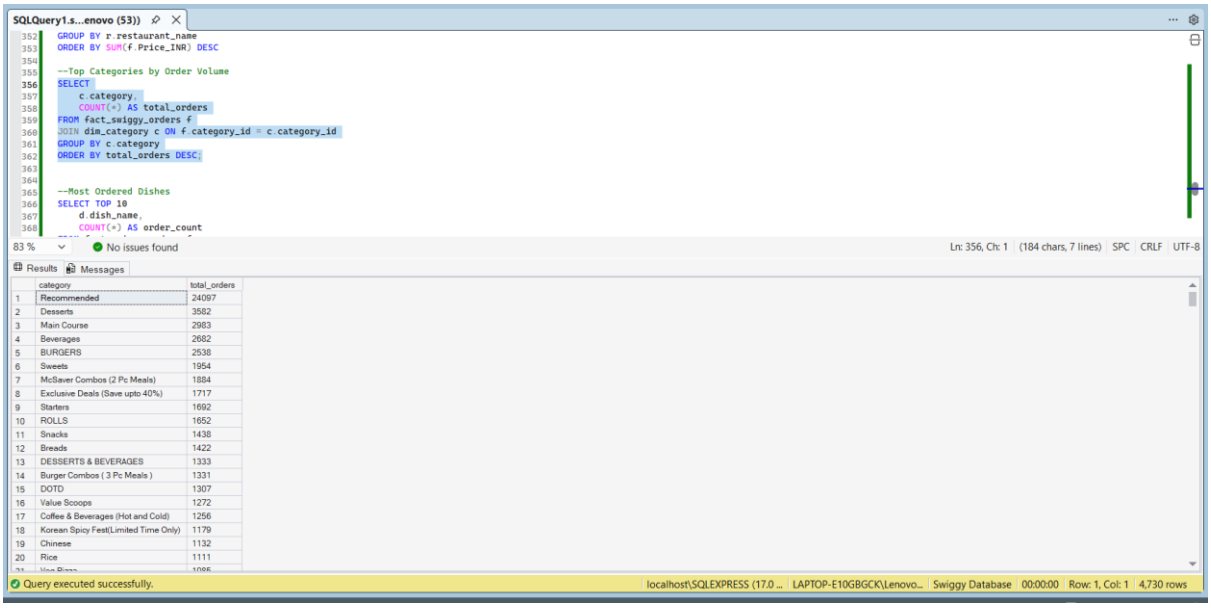
```
SELECT TOP 10
r.restaurant_name,
SUM(f.Price_INR) AS Total_Revenue FROM fact_swiggy_orders f
JOIN dim_restaurant r
ON r.restaurant_id = f.restaurant_id
GROUP BY r.restaurant_name
ORDER BY SUM(f.Price_INR) DESC
```



SWIGGY SQL SALES ANALYSIS PROJECT

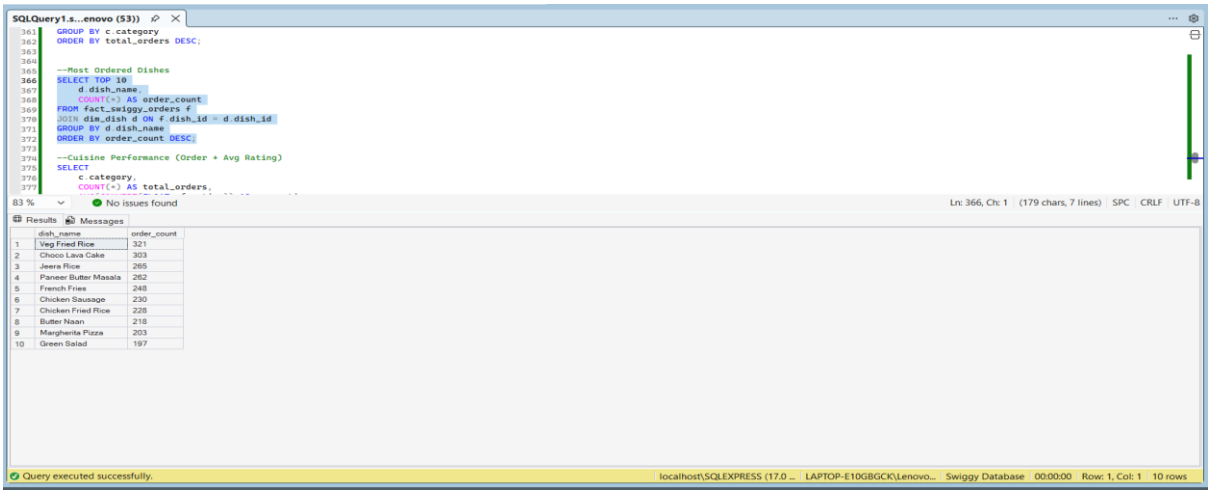
--Top Categories by Order Volume

```
SELECT
   c.category,
   COUNT(*) AS total_orders
FROM fact_swiggy_orders f
JOIN dim_category c ON f.category_id = c.category_id
GROUP BY c.category
ORDER BY total_orders DESC;
```



--Most Ordered Dishes

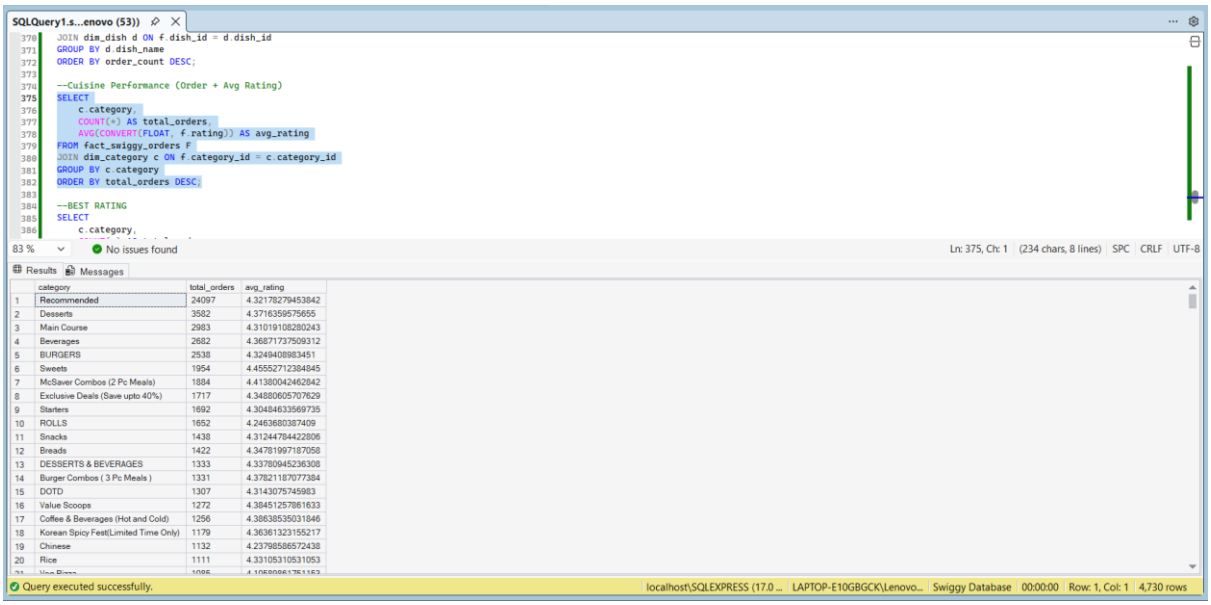
```
SELECT TOP 10
   d.dish_name,
   COUNT(*) AS order_count
FROM fact_swiggy_orders f
JOIN dim_dish d ON f.dish_id = d.dish_id
GROUP BY d.dish_name
ORDER BY order_count DESC;
```



SWIGGY SQL SALES ANALYSIS PROJECT

--Cuisine Performance (Order + Avg Rating)

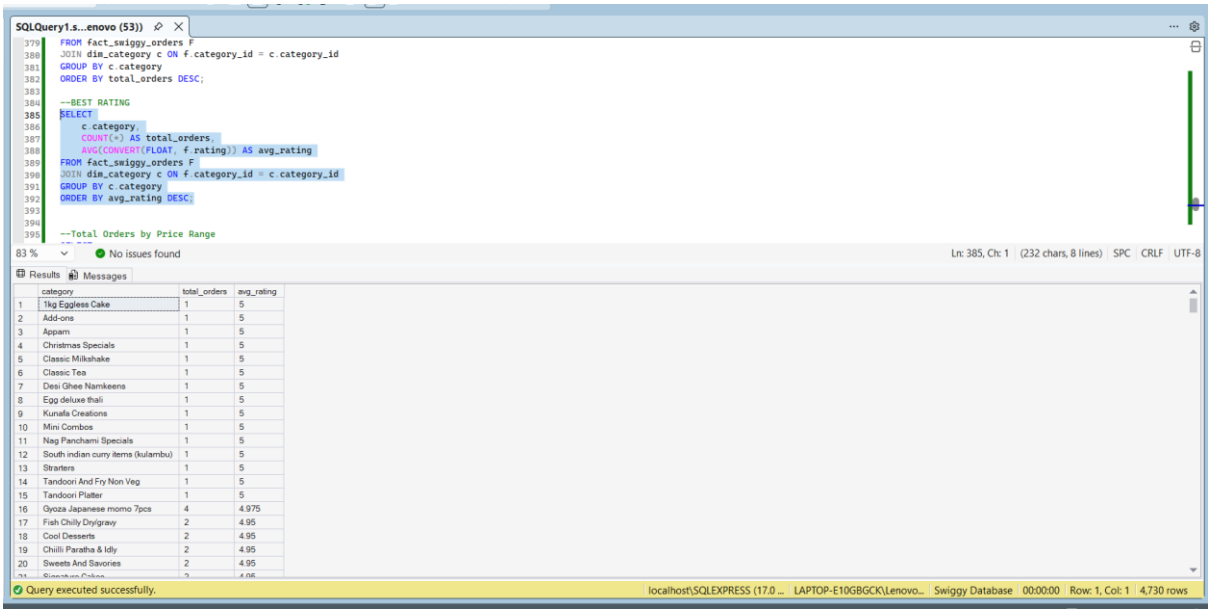
```
SELECT
    c.category,
    COUNT(*) AS total_orders,
    AVG(CONVERT(FLOAT, f.rating)) AS avg_rating
FROM fact_swiggy_orders F
JOIN dim_category c ON f.category_id = c.category_id
GROUP BY c.category
ORDER BY total_orders DESC;
```



--BEST RATING

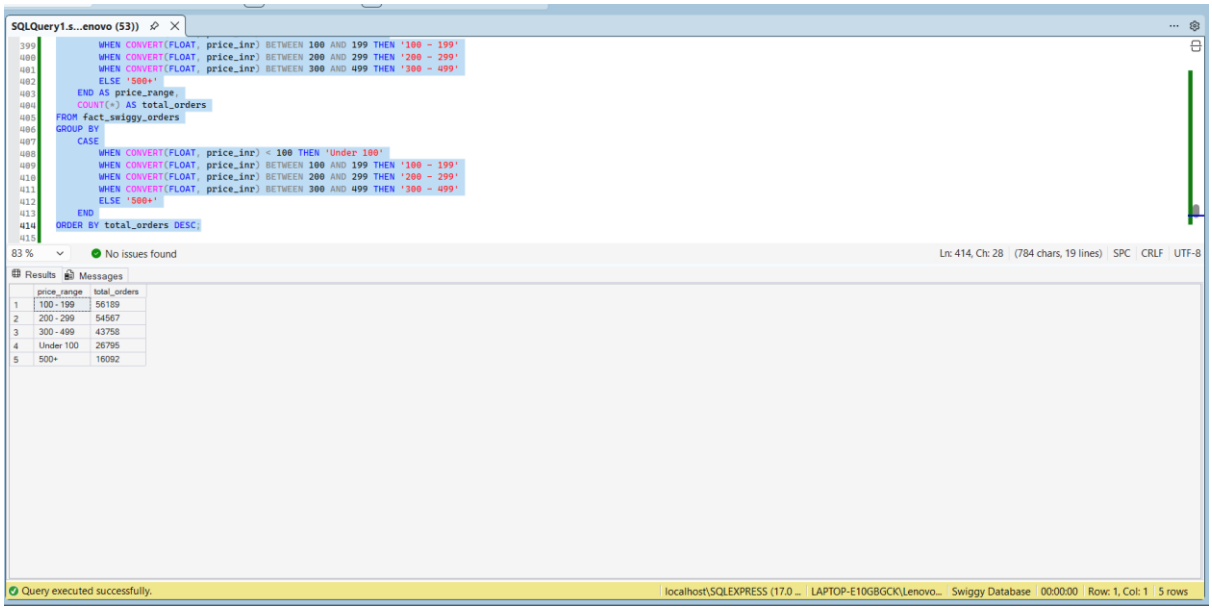
```
SELECT
    c.category,
    COUNT(*) AS total_orders,
    AVG(CONVERT(FLOAT, f.rating)) AS avg_rating
FROM fact_swiggy_orders F
JOIN dim_category c ON f.category_id = c.category_id
GROUP BY c.category
ORDER BY avg_rating DESC;
```

SWIGGY SQL SALES ANALYSIS PROJECT



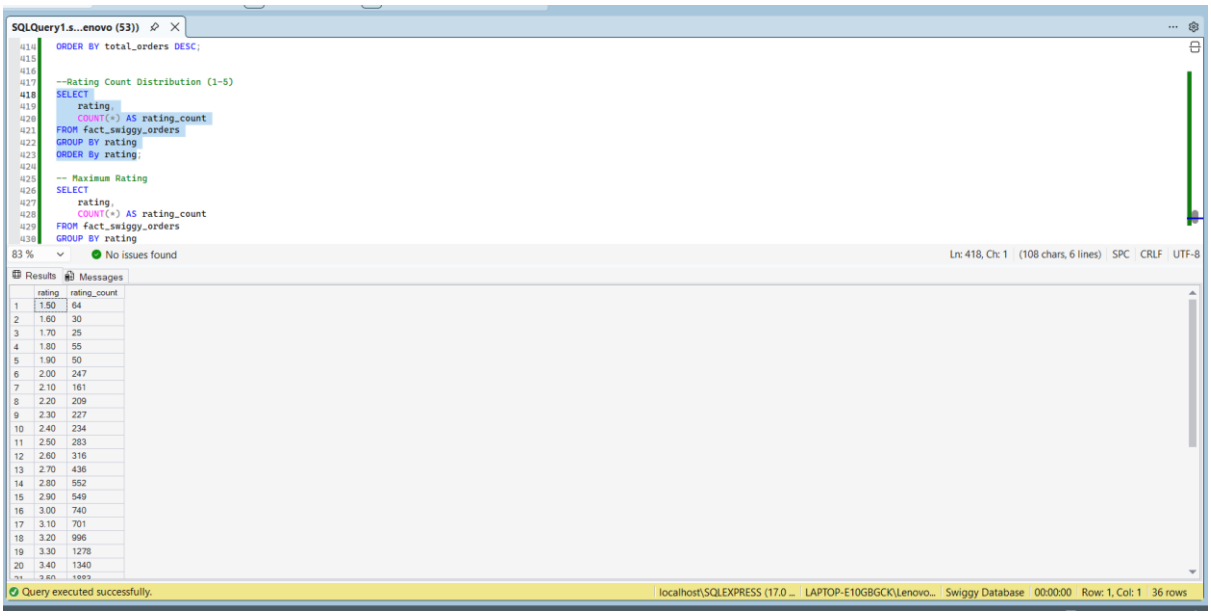

```
--Total Orders by Price Range
SELECT
   CASE
      WHEN CONVERT(FLOAT, price_inr) < 100 THEN 'Under 100'
      WHEN CONVERT(FLOAT, price_inr) BETWEEN 100 AND 199 THEN '100 - 199'
      WHEN CONVERT(FLOAT, price_inr) BETWEEN 200 AND 299 THEN '200 - 299'
      WHEN CONVERT(FLOAT, price_inr) BETWEEN 300 AND 499 THEN '300 - 499'
      ELSE '500+'
   END AS price_range,
   COUNT(*) AS total_orders
FROM fact_swiggy_orders
GROUP BY
   CASE
      WHEN CONVERT(FLOAT, price_inr) < 100 THEN 'Under 100'
      WHEN CONVERT(FLOAT, price_inr) BETWEEN 100 AND 199 THEN '100 - 199'
      WHEN CONVERT(FLOAT, price_inr) BETWEEN 200 AND 299 THEN '200 - 299'
      WHEN CONVERT(FLOAT, price_inr) BETWEEN 300 AND 499 THEN '300 - 499'
      ELSE '500+'
   END
ORDER BY total_orders DESC;
```

SWIGGY SQL SALES ANALYSIS PROJECT



--Rating Count Distribution (1-5)

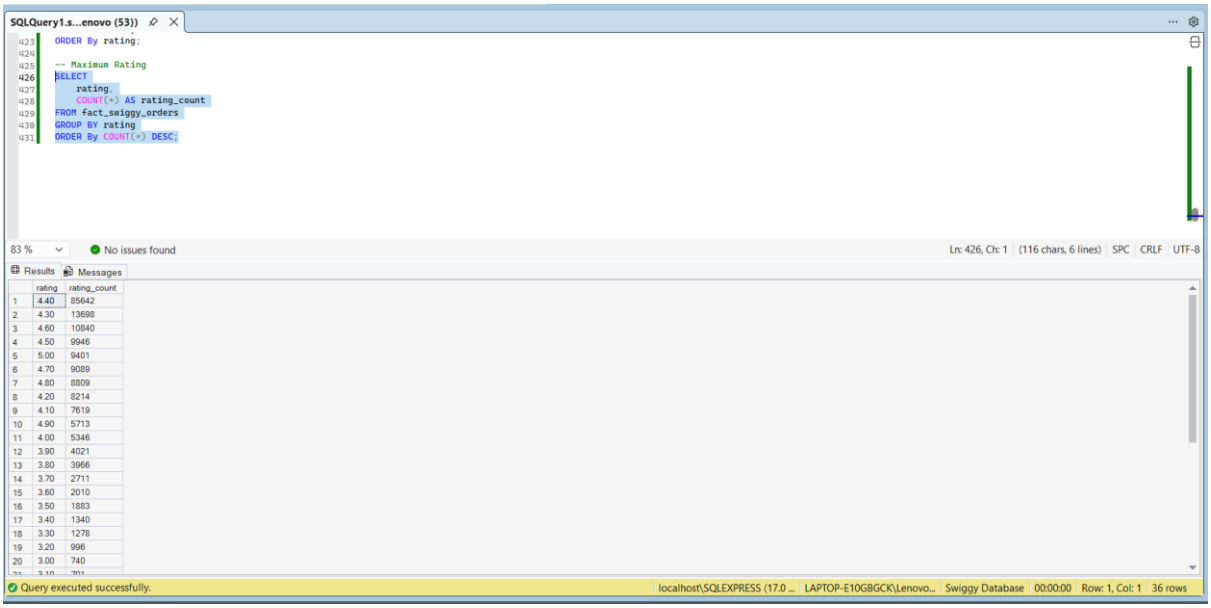
```
SELECT
    rating,
    COUNT(*) AS rating_count
FROM fact_swiggy_orders
GROUP BY rating
ORDER By rating;
```



SWIGGY SQL SALES ANALYSIS PROJECT

-- Maximum Rating

```
SELECT
   rating,
   COUNT(*) AS rating_count
FROM fact_swiggy_orders
GROUP BY rating
ORDER By COUNT(*) DESC;
```



	rating	rating_count
1	4.40	85642
2	4.30	13698
3	4.60	10840
4	4.50	9946
5	5.00	9401
6	4.70	9089
7	4.80	8809
8	4.20	8214
9	4.10	7619
10	4.90	5713
11	4.00	5346
12	3.90	4021
13	3.80	3966
14	3.70	2711
15	3.60	2010
16	3.50	1883
17	3.40	1340
18	3.30	1278
19	3.20	996
20	3.00	740